PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM TWO PRE- MOCK EXAMINATION

033 **BIOLOGY**

TIME :2:30 HOURS **MARCH 2026**

INSTRUCTIONS

- This paper consists of sections A, B and C with total of ten (10) questions.
- Answer all questions in section A, B and two questions in section C
- Section A carries fifteen (15) marks, section B carries seventy (70) and Section C carries fifteen (15) marks
- All writings must be in blue or black ink and diagrams must be drawn in pencil.
- Cellular phones and any unauthorized material are not allowed in the assessment room
- Write your name on every page of your answer sheet.

NO.	SCORE	SIGNATURE
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
TOTAL		

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CHECKER'S SIGN	

SECTION A (15 MARKS)
Answer all questions in this section.

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter on the space provided

i. The following are the basic skills in learning biology Except
   A. Measurement
   B. Experiment
   C. Hypothesis
   D. Observation

ii. Malson was given task to grid slices of Irish potatoes which apparatus should be used in this task?
   A. Knife
   B. Scissors
   C. Motor and Pestle
   D. Stones

iii. Which of the following is the proper way of positioning a light microscope on a laboratory bench or table during observation?
   A. The arm faces you while the stage faces a way from you
   B. The arm face away from you while the stage faces you
   C. The mirror looks upward
   D. The mirror looks downward

iv. A pond water drop was observed under a microscope. A unicellular organism was seen moving by using pseudopodia. Which one among the following organism represent it?
   A. Amoeba
   B. Euglena
   C. Paramecium
   D. Plasmodium

v. Which one of the following adaptations of palisade mesophyll makes them effective for photosynthesis in plant leaves?
   A. Arranged parallel to light source for maximum absorption
   B. Consist of chloroplasts with chlorophylls for capturing light.
   C. Direct connected to stomata in order to get enough carbon dioxide
   D. Located in lower epidermis to protect from light damage.

vi. Which one of the following branches of biology deals with the study of insects?

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A. Cytology
B. Ecology
C. Entomology
D. Parasitology

vii. One of the following is not a characteristic of prokaryotic cells.
A. They have flagella
B. They reproduce by multiple fussion
C. They have no true nucleus
D. They have no membrane

viii. Jad poles and fish were found in the pond. Suggest the common respiratory surface adapt their free survival in the pond.
A. Lungs
B. Moist skin
C. Tracheal system
D. Gills

ix. In scientific research, which of the following is not among procedure?
A. Experimentation
B. Hypothesis formulation
C. Identification of the problem
D. Evaluation

x. If the eyepiece magnification is 15X and the objective lens magnification is 20X. What will be the total magnification?
A. 15X
B. 300X
C. 35X
D. 20X

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the adaptation in list A with the corresponding part of the heart in B by writing the latter of the correct response beside the item number in the answer booklet provided.

LIST A	**LIST B**
i. Contracts to jump blood to body part. ii. Never get fatigue from its construction and relaxation. iii. Ensure blood flow in one direction toward the body parts. iv. Separate left and right side of the heart v. Remove waste from the whole heart.	A. Cardiac muscles B. Coronary artery C. Coronary vein D. Semilunar valves E. Septum F. Sinoatrial node G. Ventricular walls H. Connected to Aorta

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LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 MARKS)

3. **(**a) In a debate competition which involve two schools, the motion states that "Worms are non- living" Your school is opposing this motion and you are selected to be one of the main speakers to defend your school. With five (5) points can you defend your side?

   i. …….
   ii. ……
   iii. ……
   iv. ……
   v. ……

   (b) In the recent years, some secondary schools do not like to study biology as one of their subjects, on their argument that it is difficult to comprehend. Explain five (5) importance of studying this subject.

   i. …….
   ii. ……
   iii. ……
   iv. ……
   vi. ……

4. (a) You encounter several unfamiliar plants and need to classify them into respective divisions. In four (4) points, how can you distinguish Angiosperm from other plants?

   i. …….
   ii. ……
   iii. ……
   iv. ……

   (b) Write the scientific name of the following organism

   i. Lion…….
   ii. Human being……..
   iii. Maize……..
   iv. Housefly…….
   v. Domestic cat……
   vi. Mimosa plant……..

5. (a) Distinguish an owl cell from a maize cell (5 points)

An owl cell	**Maize cell**
1.	
2.	
3.	
4.	
5.	

   (b) As a competent students in cytology, state the components of cell theory as proposed by Robert Hook(5 points)

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i. …….
ii. ……
iii. ……
iv. ……
v. ……

6. (a) The process of photosynthesis require two raw material and two conditions. What are they?
   i. Raw materials……..
   ii. Conditions……

(b) Write down the chemical Equation of the process of photosynthesis

(c) Why is photosynthesis is a vital process for all living organism (4 points).

i. …….
ii. ……
iii. ……
iv. ……

7. (a) Draw a well -labelled diagram of human digestion system

(y absorbed
……..
ts functions.
i …….
i ……
iii. ……
iv. ……

8. Mkude failed to give a clear explanation of some word that are used in biology. As a form two student give two differences between.
   i. Virus and Bacteria…….
   ii. Artificial and Natural classification…….
   iii. Biology laboratory and school library…….
   iv. Holozoic and Saprophytic nutrition…….
   v. Diffusion and Osmosis……

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9. Similar types of three animal cells were placed in different solution namely A, B and C. After sometime the cell in solution A bursted, the cell in solution B shrink and shrivelled and the cell in solution C did not change.

   a) What type of solution were A, B and C
      A……
      B……...
      C. ……...
   b) Explain the reason for such observation
      a) ……
      b) ……
      c) ……

   c) Explain the main characteristic of respiratory surface (4 points)
      i. ……..
      ii. ……..
      iii. ……...
      iv. ……..

SECTION C

10. Songambele village believe that all Insects are dangerous. They decided to kill them by spraying Insecticide. As a biologist, convince them that insect are beneficial to their lives. (6 points)

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